

Technical Cybersecurity Report

Network Analysis using Nmap

Conduct a comprehensive network scan on your own IP address using Nmap, incorporating the **-sC**, **-sV**, **-vv**, **-sL**, **-sn**, and **-Pn** options. Provide a detailed report outlining the executed commands and the results obtained from the scan.)

By

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1. Executive Summary

Overview:

A comprehensive network analysis was conducted on my ip address 192.168.40.1 using Nmap as part of a routine security task. Nmap utilized various options such as; -sC (script scanning), -sV (service version detection), -vv (verbose output), -sL (list scan), -sn (ping scan), and -Pn (no ping) to identify open ports, running services, and potential vulnerabilities. The scan analysis provided a critical insight into the network device helping to gauge its exposure to threat.

Key Findings:

Open Ports Identified:

- Port 135 (Microsoft RPC)
- Port 139 (NetBIOS-SSN)
- Port 445 (Microsoft DS)
- Port 903 (VMware Authentication Daemon)
- Port 8000 (Splunkd HTTP Server)
- Port 8089 (Splunkd SSL HTTP)
- Port 16992 (Intel AMT SOAP HTTP)

Security Risks:

- Self-Signed SSL Certificate on Splunk services (Port 8089), increasing risk of man-in-the-middle (MITM) attacks.
- Splunkd HTTP Service: Exposed login page at <http://192.168.40.1:8000/en-US/account/login>, potentially vulnerable to brute-force attacks.
- The HTTP service running on port 8000 displayed no proper title and contained a **robots.txt** entry, indicating a possible misconfiguration in the Splunk instance.
- Intel AMT Management (Port 16992), which may be vulnerable to remote exploitation.

2. Background and Objectives

Project Context:

The scan analysis was conducted to scrutinize the critical network security of my IP address 192.168.40.1. Nmap was selected cause of it ability and expertise to map out network vulnerabilities and spot threats vectors

Objective of the Tool Use:

The specific objective of using Nmap was to:

- Identify open ports and running services on 192.168.40.1.

- Identify service versions to evaluate potential vulnerabilities or outdated software.
- Verify host responsiveness and analyze network behavior under various scan conditions such as **-vv**, **-sL**, **-sn**, and **-Pn**

3. Methodology

3.1 Tool Configuration

Nmap Configuration:

- Command Executed: `nmap -sC -sV -vv -sL -sn -Pn 192.168.40.1`
- Set up with **-sC** for script scanning, **-sV** for version detection, **-vv** for verbose output, **-sL** for list scanning, **-sn** for ping scanning, and **-Pn** to bypass ping checks.
- The specified target is the IP address **192.168.40.1**.

Options Used:

- **-sC** → Default script scanning to detect vulnerabilities.
- **-sV** → Service version detection.
- **-vv** → Verbose output for detailed logging.
- **-Pn** → Bypass ICMP checks, assuming the host is online.

Environment: Windows host with Nmap-Zenmap GUI

3.2 Execution Process

Network Scan Execution:

Between 13:01 and 13:13 W. Central Africa Standard Time on April 07, 2025, multiple scans were conducted to analyze the target system.

The process involved:

- A list scan (**-sL**) to validate the target.
- A ping scan (**-sn**) to determine host availability.
- A full SYN stealth scan (**-sC -sV**) incorporating script execution and version detection.
- Additionally, verbose output (**-vv**) was enabled, providing detailed logs of port discovery and service responses.

3.3 Monitoring and Analysis

System Monitoring:

- Nmap obtained TCP responses, indicating open ports and service banners.
- Scripts (-sC) scrutinized services for vulnerabilities, such as weak configurations in SMB or HTTP.

Network Monitoring:

- The scan tracked both inbound and outbound traffic to 192.168.40.1, highlighting low latency (0.00012s) and stable TTL values (128), which point to a local Windows-based host.
- No ICMP responses were detected, likely due to firewall restrictions, which were circumvented using the -Pn option.

4. Findings and Analysis

4.1 Indicators of Compromise (IOCs):

Detected IOCs:

- IP Address: 192.168.40.1
- Open Ports and Services:
 - 135/tcp: MSRPC (Microsoft Windows RPC)
 - 139/tcp: NetBIOS-SSN (Microsoft Windows NetBIOS-SSN)
 - 445/tcp: Microsoft-DS (SMB)
 - 903/tcp: VMware Authentication Daemon 1.10
 - 8000/tcp: Splunkd HTTP (http-alt)
 - 8089/tcp: Splunkd HTTPS (SSL, SplunkServerDefaultCert)
 - 16992/tcp: Intel AMT SOAP HTTP (v15.0.50.2633)
- Potential IOC: Self-signed SSL certificate on port 8089 (valid 2025-03-03 to 2028-03-02).

4.2 Behavioral Analysis

Service Actions:

- SMB Services (135, 139, 445): Enabled with message signing required (SMB2 3:1:1), typical for Windows environments but vulnerable to exploitation if misconfigured.
- Port 903 (VMware): Responded with VMware Authentication Daemon v1.10, an older version potentially susceptible to known exploits.
- Port 8000 (HTTP): Hosted a Splunk instance with a disallowed robots.txt entry (/_/) and no page title, indicating a possible misconfiguration.

- Port 8089 (HTTPS): Secured with a self-signed certificate, risking trust issues or interception.
- Port 16992 (AMT): Intel AMT service exposed, potentially allowing remote management if not properly secured.

Persistence Mechanisms:

- No direct persistence mechanisms were observed, Exposed management services, such as Intel AMT (16992/tcp) and VMware Authentication Daemon (903/tcp), present significant security concerns. If compromised, attackers could use these services to establish persistence, allowing long-term access and control over the system.

4.3 Risk and Impact Assessment

Potential Impact:

- The outdated VMware Authentication Daemon (903) could be susceptible to remote code execution, posing a risk of full system compromise
- Exposed Splunk services (8000, 8089) and Intel AMT (16992) may allow unauthorized access to management interfaces, potentially leading to data breaches or network disruptions.
- Unprotected SMB ports (139, 445) may facilitate lateral movement within the network or enable ransomware attacks, such as those exploiting WannaCry vulnerabilities.

5. Recommendations

5.1 Immediate Remediation Actions:

Service Quarantine:

- Restrict access to ports 139, 445, 903, 8000, 8089, and 16992 to authorized IPs only.
- Restrict access to Splunk login pages using IP-based firewall rules.

5.2 Long-Term Mitigation:

Strengthening Security Measures

- Upgrade VMware Authentication Daemon to the latest version to patch known vulnerabilities and enhance system security.
- Replace the self-signed certificate on port 8089 with a trusted CA-signed certificate to improve authentication and prevent interception risks.

- Strengthen SMB security by disabling outdated protocols (e.g., SMBv1) and enforcing encryption to mitigate potential exploitation.

Monitoring Enhancements

- Configure **SIEM solutions** (e.g., *Splunk alerts*) to track unauthorized login attempts.

6. Conclusion

Summary of Findings:

Nmap pinpointed seven (7) open ports on my IP 192.168.40.1 including Splunk, VMware, and Intel AMT services all of which poses a potential security risk the dated VMware Authentication Daemon and exposed management interfaces pose significant risks, potentially allowing unauthorized access or system compromise.

Next Steps:

- Restrict access to Splunk login pages using IP-based firewall rules.
- Configure **SIEM solutions** (e.g., *Splunk alerts*) to track unauthorized login attempts.
- Replace the self-signed certificate on port 8089 with a trusted CA-signed certificate to improve authentication and prevent interception risks.

Appendix: Additional DataRaw

Logs and Tool Outputs:

SampleNmap output:



